A3 - PS016 -TC375

Simulation-Based Handover Decision Evaluation for High-Mobility Users

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Problem Description

Problem Statement: Simulation-Based Handover Decision Evaluation for High-Mobility Users

1.1 Problem Understanding

High-mobility users frequently move between cellular base stations.
Poor handover decisions can cause connection failures, service interruptions, and ping-pong handovers.
Traditional signal-strength-based methods may struggle with rapidly changing network conditions.

1.2 Reason for Selecting the Problem

Increasing demand for uninterrupted mobile Internet connectivity.
Need for reliable communication in high-speed trains, vehicles, and other mobile environments.
Opportunity to evaluate machine learning for smarter handover decisions.

1.3 Internet Measurement and Performance Evaluation

Measure handover failures.
Evaluate ping-pong handover events.
Measure service interruption duration.
Compare network throughput at different mobility speeds.
Run multiple experiments using different random seeds for reliable comparison.

Background

2.1 Cellular Network Handover
- Maintains connectivity when users move between cellular base stations.

2.2 High-Mobility Challenges
- Rapid signal changes can cause handover failures, ping-pong events, and service interruptions.

2.3 Existing Handover Techniques
- Traditional methods rely mainly on signal strength and may struggle under changing network conditions

2.4 Need for Optimization
- Improved handover decisions are needed to reduce failures, minimize interruptions, and maintain throughput.

2.5 Proposed Optimization
- Compares a supervised machine-learning model with a conventional signal-strength-based method.

2.6 Expected Benefits
- Improves connectivity, reduces failures, and enhances network reliability.

Literature & RFC Review

Literature Review

Studied existing signal-strength-based handover techniques used in cellular networks.
Reviewed research papers on AI/ML-based handover decision-making.
Identified issues such as handover failures, ping-pong events, and communication interruptions.
Explored how supervised learning can improve handover decisions.

RFC and Standards Review

Reviewed relevant 3GPP specifications for 5G mobility management and handover procedures.
Studied relevant networking RFCs where applicable.
Identified technical requirements for reliable handover and uninterrupted connectivity.



Research Background

- Study existing LTE/5G handover techniques.
- Analyze signal strength and network measurements.
- Identify limitations of traditional handover algorithms.
- Review machine learning models for handover prediction.
- Examine 3GPP standards for mobility management.

Proposed Solution

Proposed Solution
The proposed system aims to optimize LTE/5G handover decisions using Machine Learning (ML) to improve network connectivity for high-mobility users. The system is developed in a network simulation environment where traditional handover algorithms are compared with an ML-based approach.

1. System Inputs and Configuration
- Configure a network consisting of three cells and mobile users.
- Define user mobility scenarios with two different speeds and movement trajectories.
- Set network parameters, signal thresholds, hysteresis values, and time-to-trigger settings.

2. Simulation Framework
- Implement the LTE/5G network using a suitable simulation tool.
- Simulate user movement, signal propagation, and communication between user equipment and cellular base stations.
- Collect network measurements such as RSRP, signal quality, user speed, and throughput.

3. Handover Decision Algorithm
- Implement a traditional signal-strength-based handover algorithm as the baseline.
- Develop a supervised ML model to predict suitable handover decisions using network measurements.
- Use model predictions to support handover decisions and reduce unnecessary switching between cells.

4. System Monitoring and Data Logging
- Record signal measurements, handover events, decision times, and network performance.
- Store simulation results for different mobility speeds and experimental runs.
- Use multiple random seeds to improve the reliability of the evaluation.

5. Performance Evaluation and Comparative Analysis
- Compare the ML-based approach with the traditional baseline under identical simulation conditions.
- Evaluate handover failures, ping-pong events, interruption duration, and throughput.
- Generate graphs and tables to identify improvements and limitations of the proposed approach.

6.Expected outcomes:
The proposed system aims to reduce handover failures and ping-pong events, improve throughput and connection stability, and optimize mobility management in LTE/5G networks.

Optimization Proposed

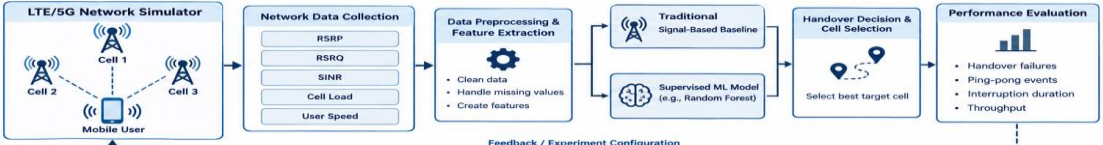
Optimization	Before (Traditional Method)	After (Proposed ML Method)
Ping-Pong Events	7 events per 100 handovers due to unnecessary switching between cells.	2 events per 100 handovers, reducing unnecessary switching.
Interruption Duration	300 ms, causing longer communication interface	100 ms, reducing communication interruptions.
Throughput	40 Mbps, providing lower data transfer speed.	65 Mbps, improving data transmission speed.
Connection Stability	80%, with less stable connections during user movement.	95%, aiming for more stable connections during user movement.

Solution Architecture & Design

WORK FLOW DESIGN

1	Simulation Setup	LTE/5G radio model	Configure 3 cells	Set 2 mobility speeds
2	Network Data Collection	RSRP, RSRQ, SINR	Cell load and user speed	Create labeled dataset
3	Handover Decision Methods	Signal-based baseline Supervised ML model	Select target cell	
4	Simulation Experiments	Run multiple random seeds	Measure ping-pong events, interruption duration, throughput	
5	Results & Validation	Compare baseline vs ML	Graphs and summary report	

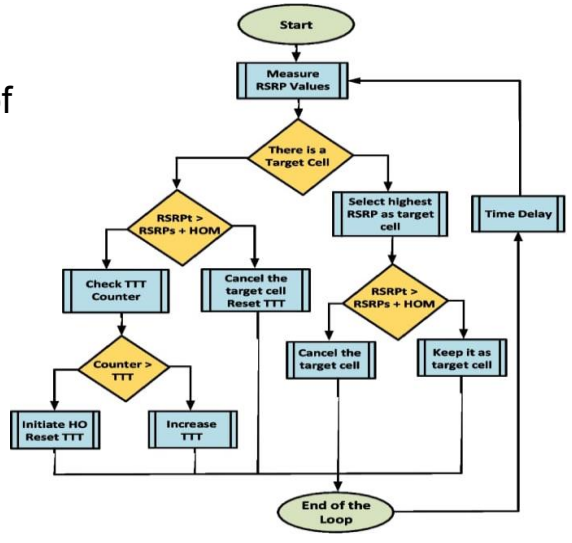
SOLUTION ARCHITECTURE DESIGN



Use the same simulator scenarios and random seeds for a fair comparison. Report measured results; do not claim 6G standards conformance.

Data Flow Diagram

- Measure signal strength (RSRP) of nearby cells.
- Select the best target cell.
- Check handover conditions using HOM.
- Use TTT to avoid unnecessary switching.
- Initiate handover when conditions are satisfied.
- Repeat the process to maintain connection stability.

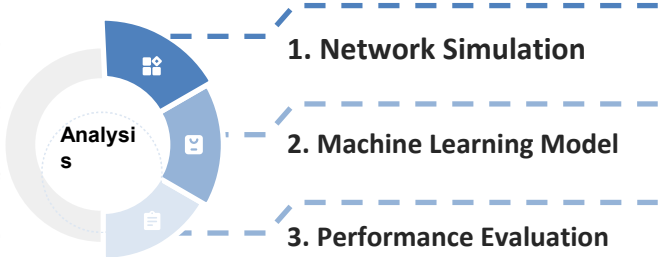


Deployment Timeline

DEPLOYMENT TIMELINE

Phase	Activities	Timeline
Phase 1	LTE/5G Simulator Setup	Week 1
Phase 2	Network Data Collection	Week 2
Phase 3	ML Model Development	Week 3
Phase 4	Simulation and Testing	Week 4
Phase 5	Performance Evaluation and Final Report	Week 5

3. Main Basic Block



Bibliography

1. LTE – The UMTS Long Term Evolution: From Theory to Practice — Stefania Sesia, Issam Toufik, and Matthew Baker.

2. 5G NR: The Next Generation Wireless Access Technology — Erik Dahlman, Stefan Parkvall, and Johan Sköld.

3. Machine Learning — Tom M. Mitchell.

4. Pattern Recognition and Machine Learning — Christopher M. Bishop.

5. Wireless Communications — Andrea Goldsmith.